

Kundan Kumar

SENIOR RESEARCH FELLOW, ELECTRICAL ENGINEERING, IIT PATNA

D-404, Boys Hostel
IIT Patna
Patna, Bihar, India
kundan.pee16@iitp.ac.in | kkr2k12@gmail.com
Researchgate : [researchgate.net/profile/Kundan-Kumar-20](https://www.researchgate.net/profile/Kundan-Kumar-20)
Linkedin : [linkedin.com/in/kkr-kundan/](https://www.linkedin.com/in/kkr-kundan/)
+91-95-466-72280

EDUCATION	Indian Institute of Technology Patna, Bihar, India Ph.D in Electrical Engineering	June 2022 (Expected)
	Muzaffarpur Institute of Technology Muzaffarpur, Bihar, India B.Tech in Electrical Engineering	July 2016
RESEARCH INTERESTS	Nonlinear filtering, State estimation, Kalman filtering, Target tracking, Bearings-only tracking, Doppler bearing tracking, Interacting multiple model filtering	
PUBLICATIONS	Journal Articles	
	K. Kumar, S. Bhaumik, and Paresh Date, "Extended Kalman Filter using Orthogonal Polynomials," <i>IEEE Access</i> , vol. 9, pp. 59675–59691, 2021.	
	K. Kumar, S. Bhaumik, and S. Arulampalam, "Tracking a Torpedo with Unknown Sensor Noise Covariance using Orthogonal Polynomial Filters," under review in <i>IEEE Transactions on Aerospace and Electronic Systems</i> .	
	K. Kumar, S. Das and S. Bhaumik, "State Estimation using Gaussian Integral," communicated to <i>Digital Signal Processing</i> .	
	K. Kumar and S. Bhaumik, "Polynomial Chaos Approximation Based Kalman Filter," (ready to submit).	
	K. Kumar and S. Bhaumik, "Remote Nonlinear State Estimation with Event Triggered Mechanism for Arbitrary-step Delayed Measurements," (ready to submit).	
	Conference Proceedings	
	K. Kumar and S. Bhaumik, "Higher Degree Cubature Quadrature Kalman Filter for Randomly Delayed Measurements," <i>21st International Conference on Information Fusion (FUSION)</i> , July 2018 (University of Cambridge), Cambridge, United Kingdom.	
	A. K. Singh, K. Kumar, and S. Bhaumik, "Cubature and Quadrature Based Continuous-Discrete Filters for Maneuvering Target Tracking," <i>21st International Conference on Information Fusion (FUSION)</i> , July 2018 (University of Cambridge), Cambridge, United Kingdom.	
	K. Kumar and S. Bhaumik, "Nonlinear Filter for a System with Randomly Delayed Measurements and Inputs," <i>Engineering Proceedings</i> , vol. 2, p. 40, Multidisciplinary Digital Publishing Institute, 2020.	
	A. K. Singh, S. Bhaumik, and K. Kumar, "Cubature quadrature filter for one-step randomly delayed measurements," <i>Indian Control Conference (ICC)</i> , pp. 36-41, January 2018 (IIT Kanpur, India).	
	S. Das, K. Kumar, R. Radhakrishnan, and S. Bhaumik, "Bearing-only tracking using range-parameterized shifted Rayleigh filter." to appear in <i>Proceedings of Ocean 2022, Chennai, India</i> .	

**RESEARCH
EXPERIANCE****Doctoral Research Experience***Supervisor : Dr. Shovan Bhaumik**July 2016 - Present*

- Developed a higher degree cubature quadrature Kalman filter (HDCQKF) for a nonlinear system with arbitrary step randomly delayed measurements
- Development of cubature quadrature Kalman filter for randomly delayed measurement and inputs
- Developed a new extended Kalman filter using orthogonal polynomial
- Proposed a filtering algorithm based on polynomial chaos expansion
- Developed a new filtering technique based on Gaussian integral
- Application of orthogonal polynomial filter in real life torpedo tracking problem in the absence of sensor noise covariance
- Developed a Bayesian filtering framework under an event-triggered mechanism for the remote state estimation in the presence of arbitrary step delayed measurement

Experience in Research Guidance at IIT Patna*Since Jan 2021***Mentored Junior Research Fellow**

- Development of range parameterized shifted Rayleigh filter for a real life BOT problem
- Developed a range parameterized adaptive filter for a real life BOT problem

**TEACHING
EXPERIENCES****Teaching Assistance at IIT Patna***July 2016 – July 2021*

- EE371 Instrumentation & Control and Laboratory *Jan-May, 2018*
- EE103 Basic Electronics Laboratory *July-Dec, 2018*
- EE371 Instrumentation & Control and Laboratory *Jan-May, 2019*
- EE103 Basic Electronics Laboratory *July-Dec, 2019*
- EE371 Instrumentation & Control and Laboratory *Jan-May, 2020*
- Mid and End-Semester Examination duties *July 2016 - July 2021*

**AWARDS &
ACHIEVEMENTS**

Three year **Senior Research fellowship (SRF)** at IIT Patna
Ministry of Human Resource Development, Government of India.

Sep 2018

Travel grant to attend **21st International Conference on Information Fusion (FUSION)**, University of Cambridge, United Kingdom

July 2018

Travel grant to attend **2018 Indian Control Conference (ICC)**,
IIT Kanpur, India

Jan 2018

Two year **Junior Research fellowship (JRF)** at IIT Patna
Ministry of Human Resource Development, Government of India.

July 2016

Graduate Aptitude Test in Engineering (**GATE**)
All India Rank: 2033 out of 125859 candidates

2016

PROFESSIONAL ACTIVITIES

IEEE Student Member, IEEE Membership Number: 97409746
Reviewer of IEEE Transactions on Circuits and Systems I: Regular Papers
Reviewer of International Conference on Control, Decision and
 Information Technologies (CODIT)
Reviewer of International Symposium of Asian Control Association
 on Intelligent Robotics and Industrial Automation (IRIA)
Organizer of Research Scholar's Day (IIT Patna)

Since 2020

2018, 2019

SKILLS

Languages: C, Python
Software: Matlab, Latex, gnuplot, MS office
Teaching: Conducted B.Tech classes at IIT Patna

REFERENCES

Dr. Shovan Bhaumik
Associate Professor (PhD supervisor)
Department of Electrical Engineering
Indian Institute of Technology Patna
Bihar, India
Email: shovan.bhaumik@iitp.ac.in
Phone: +91-612-302 8049
Mobile: +91-9709300163
Fax: +91-612-227 7383

Dr. Paresh Date
Reader & Director of Research and Head of Financial Mathematics
and Operational Research (FOR) group,
Department of Mathematics,
College of Engineering, Design and Physical Sciences,
Brunel University
Middlesex UB8 3PH
United Kingdom
Email: paresh.date@brunel.ac.uk
Phone: +44-1895-265613
Fax: +44-1895-269732

Dr. Nutan Kumar Tomar
Associate Professor (Doctoral Committee Member)
Department of Mathematics
Indian Institute of Technology Patna
Bihar, India
Email: nktomar@iitp.ac.in
Phone: +91-612-255 2021
Mobile: +91-9470836610
Fax: +91-612-227 7383